Do Now · retrieval

Algorithms: synoptic exam challenge

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

True remaining cost is 4; is h(A)=7 admissible?

Q2

If g=3 and h=4, what is f?

Q3

What condition is needed for Dijkstra's standard shortest-path guarantee?

Q4

When partition sets the pivot at index 2, which recursive index ranges remain within indexes 0–4?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Match the algorithm to the task — Quicksort partitions an array. Dijkstra finds shortest routes on non-negative weighted graphs. A* adds a remaining-cost estimate.

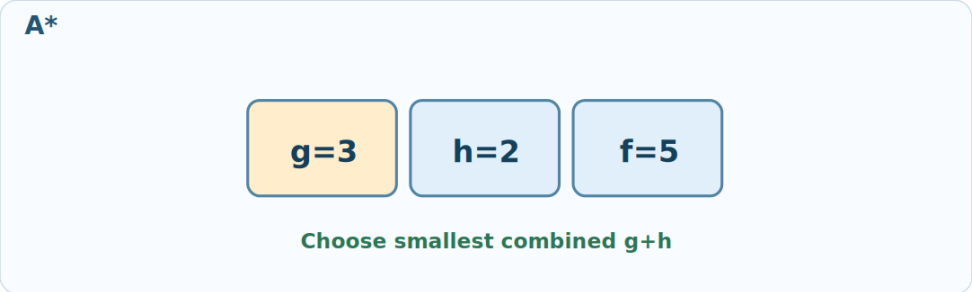
Core idea 2

Show all required intermediate states — For quicksort write arrays after partition. For shortest paths maintain distances, predecessors and a clear visited or frontier order.

Core idea 3

Explain efficiency and preconditions — State when each algorithm is appropriate and how its method changes work.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

Exam command words matter — Trace = show states. Explain = give a cause or reason. Compare = connect similarities and differences with evidence.

Vocabulary · use the correct technical terms

partition — Rearrange around a pivot and place that pivot.

tentative distance — Current best-known cost from the start.

predecessor — Vertex immediately before another on a path.

heuristic — Estimate of remaining distance to the goal.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

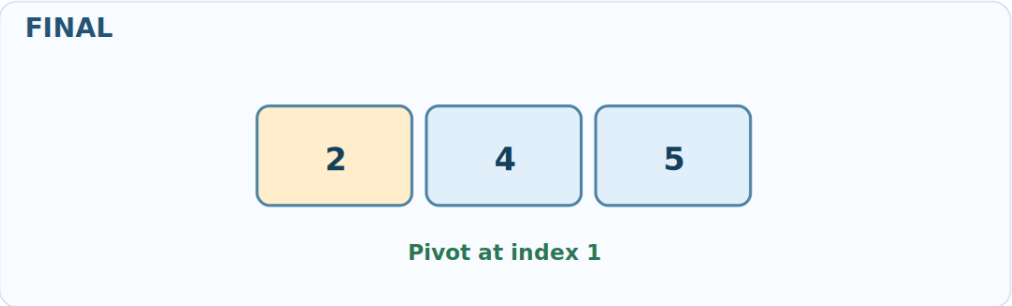
Problem

Use last-element Lomuto partition on [5,2,4]. Give final pivot index.

Follow the worked method

1. Predict first — Start with pivot 4; 5 is too large to move to the smaller section.
2. Model response — Move 2 to the left: [2,5,4].
3. Model response — Swap pivot into index 1: [2,4,5]. Only one partition is being demonstrated.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
From P, Q gets 2 and R gets 5; choose Q.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Using last-element Lomuto quicksort on [7,2,6,1,5]: (a) state initial pivot [1]; (b) show the array after moving each qualifying value [3]; (c) give array and pivot index after partition [2]; (d) give fully sorted array [2]; (e) state typical average and worst-case time complexity [2].

Question 2

Find the shortest route A→E for non-negative edges: A–B=2, A–C=6, B–C=1, B–D=7, C–D=2, D–E=3, C–E=9. Give tentative-distance improvements [8] and final path [2].

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Graph edges S–A=2, S–B=5, A–B=1, B–G=3, A–G=7. Heuristics h(S)=5,h(A)=3,h(B)=2,h(G)=0. (a) Define g, h, f [3]; (b) show initial f for A and B [2]; (c) update B after expanding A [2]; (d) state shortest route and cost [2]; (e) evaluate h(A) admissibility [1].

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Quicksort, Dijkstra and A* are interchangeable because they all have a loop.

Use key vocabulary from memory

1. Explain partition in your own words.

2. Explain tentative distance in your own words.

3. Explain predecessor in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.